Explainable AI – An Introduction

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The Black-Box Problem

Key Problem

Modern machine learning prioritizes predictive performance over transparency.

Complex algorithms achieve outstanding accuracy but hide their decision logic inside millions of parameters.

WHY INTERPRETABILITY?

In high-stakes applications, unexplainable model failure poses critical risks:

- **Medical Diagnosis (life safety)**
- **Credit Scoring (financial loss)**
- **Autonomous Driving (legal liability)**

Key Goals of Explainable AI (XAI)

01

TRUST

Verifying that predictions rely on legitimate domain logic rather than spurious correlations.

02

DEBUGGING

Identifying where, how, and why a model makes systematic errors on edge cases.

03

FAIRNESS & AUDITING

Ensuring decisions do not discriminate based on protected attributes.

The Accuracy vs. Interpretability Trade-Off

Model Type	Examples	Interpretability	Predictive Power
Glass-Box (Intrinsic)	Linear Regression, Decision Trees, Logistic Regression	**Very High**	**Moderate**
Ensemble Models	Random Forest, XGBoost, LightGBM	**Moderate**	**High**
Deep Learning	Convolutional Nets, Transformers	**Low**	**Very High**

Intrinsic Interpretability

LINEAR & LOGISTIC REGRESSION

Direct Mathematical Transparency

Each feature coefficient directly represents the expected change in outcome per unit increase in feature value.

DECISION TREES

Explicit Human Logic Rules

Decision rules follow explicit IF-THEN conditional splits that map directly to human logic.

Scopes of Explanation: Global vs. Local

GLOBAL INTERPRETABILITY

Overall Dataset Model Logic

Understanding overall logic and behavior across the entire dataset.

Key Question: "Which features drive predictions overall?"

LOCAL INTERPRETABILITY

Single Prediction Explanations

Explaining why the model made a specific decision for one individual data point.

Key Question: "Why was this specific loan application rejected?"

Permutation Feature Importance

CORE CONCEPT

Measures how much model performance decreases when a feature's values are randomly shuffled.

INTUITION

- **High Impact:**
If shuffling breaks model accuracy, the model relied heavily on that feature.

- **Low Impact:**
If performance stays unchanged, the feature was largely ignored.

LIME: Local Interpretable Model-agnostic Explanations

HOW LIME WORKS

LIME builds a simple, interpretable model (like linear regression) that mirrors the complex model's behavior only in the immediate neighborhood of a single prediction point.

GEOMETRIC ANALOGY

Imagine a complex topological map with winding curves.

If you zoom in extremely close to one specific point, the surface looks flat. LIME draws a straight line across that local flat surface.

SHAP: Game-Theoretic Interpretability

SHAP CONCEPT

SHAP adapts Shapley Values from cooperative game theory to measure feature contribution.

GAME THEORY ANALOGY

- **Prediction = Winning Team Cash Prize**
- **Features = Individual Team Players**

SHAP calculates the exact reward each player deserves by testing all possible sub-coalitions of players with and without that feature.

SHAP: Key Axiomatic Properties

1. CONSISTENCY

If a model changes so that a feature's contribution increases or stays the same, its SHAP value will not decrease.

2. ADDITIVITY

The sum of feature contributions equals the difference between the actual prediction and the average baseline prediction.

Ethical foundations of (Gen)AI

Integrity

Uphold academic honesty and scholarly standards. Never use AI to misrepresent authorship or fabricate information.

Originality

AI must support, not replace, human intellect. Core scholarly outputs must reflect individual knowledge and critical analysis.

Transparency

Clearly acknowledge and document all AI tool usage across proposals, theses, publications, and presentations.

Accountability

Users remain fully responsible for outputs and conclusions. AI reliance does not diminish individual accountability.

Compliance

Comply with legal, regulatory, and institutional requirements, including data protection laws and ethics approvals.

Security

Safeguard data confidentiality and IP. Exercise caution with third-party platforms to prevent unauthorized leaks.

Intellectual Property

Respect third-party IP rights and ensure AI tool licensing terms align with institutional policies.

Fairness & Bias

Critically evaluate AI outputs to recognize algorithmic bias and prevent unfair or discriminatory outcomes.

Human Oversight

Maintain meaningful human judgment in all activities, ensuring final decisions are made by qualified individuals.

Assignment

- **Refer:** https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=OJ:L_202401689
- **Task:** Prepare a 5-page report on the following
    1. Key concerns raised
    2. How the concerns are addressed
    3. What additional guardrails/regulations will be required and why.